

- 6 (a) (i) From $V = IR$, $I = \frac{V}{R}$ hence $I \propto \frac{1}{R}$ C1
- Ratio = 1 : 0.5 : 0.2 or 10 : 5 : 2 A1
- (ii) 1. $Q = Ne = 1.0 \times 10^{-3} \times 6.02 \times 10^{23} \times 1.6 \times 10^{-19} = 96.32 \approx 96 \text{ C (shown)}$ A1
2. Current through the resistor: $I = \frac{Q}{t} = \frac{96}{320} = 0.30 \text{ A}$ M1
- Energy dissipated = $I^2 R t = (0.30)^2 (4.0) (320)$ M1
 $= 115.2 \approx 115 \text{ J}$
- OR
- p.d. $V = IR = 0.30 \times 4.0 = 1.2 \text{ V}$ (M1)
- Energy = $QV = 96 \times 1.2$ or Energy = $\frac{V^2}{R} t = \frac{1.2^2}{4.0} \times 320$ (M1)
3. Let the current through R be I. Current through 4-ohm resistor = 3I
 Since resistors are in parallel: $I \times R = 3I \times 4$ M1
- $R = 12 \Omega$ A1
- (b) (i) 0.2 A A1
- (ii) 1.0 V A1

7

(a)

6

6

5 ohms

A1

A1